

Ideation Phase
Brainstorm & Idea Prioritization

Date	7 March 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization – Malaria Detection System

1 Clarify (Understand the Problem)

Problem Statement

Malaria diagnosis is traditionally done using microscopic examination of blood smear samples by trained technicians. This process is **time-consuming, requires expertise, and may delay diagnosis**, especially in rural or resource-limited healthcare centers.

Goal

Develop an **AI-based malaria detection system** that can analyze blood smear images and automatically identify whether malaria parasites are present.

Key Questions

- How can we detect malaria quickly using technology?
- How can AI help doctors analyze blood cell images?
- How can we make the system easy to use for hospitals and labs?

❏ Create (Generate Ideas)

During brainstorming, the team generates multiple ideas to solve the problem.

Possible Solutions

Idea	Description
AI Image Detection	Use a CNN model to detect malaria parasites from blood smear images
Web Application	Build a web platform where users can upload images
Automated Report	Generate automatic diagnosis reports
Cloud Storage	Store detection results in a cloud database
Mobile Integration	Allow doctors to upload images through mobile phones

Tools & Technologies

- Python
- TensorFlow / Keras
- OpenCV
- HTML, CSS, JavaScript
- MySQL / Firebase

Step 3 **Commit (Select Best Ideas)**

After evaluating ideas, the team prioritizes the most effective solutions.

Selected Solution

Develop a **Malaria Detection System using Deep Learning** that allows users to:

1. Upload microscopic blood smear images
2. Process the image using OpenCV
3. Analyze the image using a CNN model
4. Display results as **Parasitized or Uninfected**
5. Store results for future medical reference

Expected Benefits

- Faster malaria diagnosis
- Reduced dependency on manual microscopy
- Assist doctors in early detection
- Improve healthcare accessibility